

Table of Contents

PACKING, LABELLING AND SHIPPING

Document: PKG-EEG-015 **Revision:** B **Date:** 2026-09-01 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Licence:** CC BY-SA 4.0
Governing documents: DSN-EEG-003 Rev D, then RFQ-EEG-001 Rev E. Where this document and design.py disagree, design.py governs.

Revision letters used below. Every cross-reference in this document names the current issue of the document it cites: DSN-EEG-002 Rev E, DSN-EEG-003 Rev D, RFQ-EEG-001 Rev E, FW-EEG-001 Rev D, TST-EEG-004 Rev C, SCH-EEG-005 Rev B, ICD-EEG-006 Rev C, ASM-EEG-007 Rev B, WH-EEG-008 Rev B, JIG-EEG-009 Rev C, QP-EEG-010 Rev B, RISK-EEG-011 Rev B, REG-EEG-012 Rev B, SVC-EEG-013 Rev B, IFU-EEG-014 Rev B, ECO-EEG-016 Rev C, AVL-EEG-017 Rev C, SIM-EEG-018 Rev A, PARTS-EEG-019 Rev B, MECH-EEG-020 Rev A and RUL-EEG-021 Rev B. Where an earlier letter appears it is deliberate and names a superseded issue, as in the Rev C and Rev D readings of RFQ M-05 in section 3.1.

Rev B in one line: the carrier is now 150.0 × 130.0 mm on four layers and the pod grew with it to 163.0 × 143.0 × 62.0 mm closed, so every packed dimension and every calculated mass is restated; the serial format becomes TI0V-B-nnnn and section 5 of this document is stated to be its single home; the headband is withdrawn from the packing list and RFQ A-03 is met by the chin strap and the occipital yoke; the calibration certificate moves to the case lid pocket; the invented hatch interlock is deleted; the travel-case internal dimensions are settled in section 3 on one transcription of one shell; and the foam schedule is now checked against part **depth** as well as footprint, which CASE-00 Rev B fails everywhere, so section 2.2 carries a Rev C schedule drawn to the settled case.

Closed in this issue: the findings raised against this document by the second cross-document audit of 1 September 2026 – 2.2 on where the serial format lives, 2.7 on the travel-case internals, and 4.1 on governing-document revision letters. These are corrections within the Rev B release and the revision letter is not advanced for them.

Corrections after the verification review of package v2.2 (2026-09-02), revision letter again unchanged. The kit packing list now carries **all five** of the printed items IFU-EEG-014 Rev B section 14 puts in the case lid wallet: the placement guide and the full IFU are added as lines 6.2 and 6.3 and group 6 is renumbered, which closes IFU-EEG-014 section 16 item 14, and sections 2.2, 2.3 and the lid row of section 3.2 count five where they counted three. Section 1.1 gains a check of every pocket name against the seven CASE-00 Rev C cut files. The two citations of **DSN-EEG-002 Rev E section 10** in section 1.2 are marked as pointing at a superseded section – that table has no HM-10, still calls CASE-00 a two-layer insert, and reads HM-04 as 8 + 2 spare per kit – because DSN-EEG-002 has not been re-issued and RISK-EEG-011 Rev B section 6.1 records that it must be corrected. And the state-of-charge row of section 7 and open item 8 now say plainly that open item 8 is the **only** register entry the package holds for the missing fuel-gauge step, because RFQ-EEG-001 Rev E section 9.1 item 12 raises it “as an open item” against a register that carries no such item.

The label and packaging artwork is generated, at this issue, revision letter again unchanged. ECO-EEG-016 Rev C section 1 has registered ART-LBL-01 to ART-LBL-07, ART-PACK-01, ART-DIS-01, ART-RET-01 and DRW-LBL-PLACEMENT as generated artifacts under graphics/ since Rev B was issued, and none of them existed. Section 1.1 ticked against three files that were not in the package, section 4.2 made the QA placement sign-off a check against a drawing that was not in the package, and the nine foam bay tags could not be ordered as artwork at all. tools/artwork_gen.py now writes all eleven into graphics/labels/, reading every legend, size, substrate and piece of copy out of this document rather than out of a person's retyping. Section 4.3 is the index. Generating them found three things this document had wrong or unstated, and they are open items 14, 15 and 16 rather than silent corrections: the regulated lithium pictogram is not in this package and is not drawn; the 60 x 12 mm bay tag cannot carry the 8 mm cap height section 2.3 asks for, on any of the nine legends; and the spare-cell tag is specified one way in section 4.2 and another in section 7.

Why this document exists

The v1 package told a manufacturer to tick a packing list that did not exist, to cut a foam insert from one DXF with no thickness and no bottom layer, to apply a label whose serial format was never defined, to buy a travel case that three documents sized three different ways, and to post a lithium cell to a stranger and get it back without once naming a UN number. This document is the packing list, the foam and case specification, the marking set, the serial-number scheme, the carton specification and the dangerous-goods procedure, in one place, so that final assembly step T18 of TST-EEG-004 Rev C and step 10 of RFQ-EEG-001 Rev E section 9.1 can actually be performed and signed. It closes the v1 audit entries kit-packing-list, foam-insert-layout-drawing, mech-foam-dxf-defects, case-selection-and-fit-spec, foam-labelling-and-packing-photograph, unit-label-artwork-and-serial-scheme, case-and-carton-marking-set, mech-label-artwork, outer-carton-and-return-shipping, lithium-shipping-procedure, lithium-shipping-dossier and mech-packing-and-shipping, and the serialisation half of serialisation-and-dhr.

Nothing in this package has been manufactured, packed, weighed or shipped, and **no safety engineer has reviewed this design**. Every mass, every carton dimension and every chargeable-weight figure below is **calculated** from part geometry and stated densities, and is marked as such. Section 2.4 records, plainly, why the two-sheet CASE-00 Rev B insert could not pack the kit, on footprint and on depth. The seven Rev C cut files that replace it are released with this issue and the Rev B pair is deleted; what is still not settled is the sheet size, because it is drawn to a case internal dimension that nobody has measured.

What changed in the design, and what it does to the packing

Two things changed during layout. Both are engineering findings rather than preferences, and both land on this document, because packing is where board geometry finally becomes freight cost.

The carrier grew from 130 × 124 mm to 150.0 × 130.0 mm. Thirty connectors, 211 parts and 156 nets would not close at the smaller size. At kit quantities the extra 33.8 cm² of bare board costs a few euro per unit against a real risk of an unroutable design.

The carrier went from two layers to four: L1 signal, L2 reference plane, L3 reference plane, L4 signal. Package v1 asserted that a two-layer carrier would be cheap and easy to route. Actually laying it out showed that it is not: on two layers the bottom side has to

be both the reference plane and the second routing surface, and it cannot be both. Four layers give two full routing surfaces and a continuous reference under every analogue trace, which is what DSN-EEG-002 Rev E section 13's "layout rules that are requirements, not preferences" ask for, and which a swiss-cheesed two-layer pour cannot deliver. The zoning, star-point and isolation rules that go with them are DSN-EEG-003 Rev D section 3.3. At 2 units that is about €35 more in total; at 50 units it is about €3 per board. The reference planes are AGND_REF left of x = 62 mm and DGND right of it, on both inner layers, with through vias only, 0.60 mm pad on a 0.30 mm finished hole, in a stack of mask / 35 µm L1 / prepreg 0.200 / 17 µm L2 / core 1.065 / 17 µm L3 / prepreg 0.200 / 35 µm L4 / mask = 1.60 mm ± 10 %.

The enclosure grew with the board. POD-P1 is now **163.0 × 143.0 × 58.0 mm external and 158.0 × 138.0 × 55.5 mm internal**, with a lid of 163.0 × 143.0 × 6.0 mm carrying a 2.0 mm spigot, so the pod as packed is **163.0 × 143.0 × 62.0 mm closed**. The MP-01 module plate is 146.0 × 126.0 × 3.0 mm. The consequences for this document are not cosmetic: the POD-P1 ENCLOSURE foam pocket is now short in both axes and in depth, the pod assembly is about 130 g heavier, the foam sheet has to be re-cut, and both the case and the carton grow.

1. Kit packing list (KPL-EEG-001 Rev B)

This table is KPL-EEG-001. TST-EEG-004 Rev C T18 and RFQ-EEG-001 Rev E section 9.1 step 10 tick against it, line by line. It is derived from EEG_kit_BOM_for_bidders_RevB and from the part register PARTS-EEG-019 Rev B and DSN-EEG-003 Rev D section 4; where the kit BOM was wrong, the corrected quantity is given here and the correction is listed in section 1.2.

Pocket names are the legends of the foam schedule in section 2.2. "Fitted" means the item is already assembled into the helmet or the pod and is ticked in place, not packed loose. Part identifiers are those of PARTS-EEG-019, which owns them.

1.1 The list

#	Part	Description	Qty	Pocket	Serialised	Tick
1 INSTRUMENT						
1.1	HM-01	Helmet frame monocoque, MJF PA12, with channel cover strips	1	HELMET HM-01	yes (kit id)	☐
1.2	HM-04	Electrode assemblies bonded into HM-01	8 fitted	HELMET HM-01	no	☐
1.3	HM-05	Sintered Ag/AgCl cups on service bayonets, fitted	8 fitted	HELMET HM-01	no	☐
1.4	HM-05	Sintered Ag/AgCl cups, spare	2	SPARE CUPS + KEYLESS SPARES	no	☐
1.5	HM-02	TPU 85A pads: brow, occiput ×2, crown, fitted	4 fitted	HELMET HM-01	no	☐
1.6	HM-02	TPU 85A pads, spare set	4	SPARE CUPS + KEYLESS SPARES	no	☐

#	Part	Description	Qty	Pocket	Serialised	Tick
1.7	HM-03	Occipital yoke, ratchet dial housing and bought-in ratchet (HM-03A/B/C), fitted. This and line 1.8 are what satisfies RFQ A-03	1 fitted	HELMET HM-01	no	☐
1.8	HM-06	Chin strap, chin cup and liner (HM-06A/B/C), fitted. With line 1.7, satisfies RFQ A-03	1 fitted	HELMET HM-01	no	☐
1.9	HM-06B	Chin-cup liner, spare	1	SPARE CUPS + KEYLESS SPARES	no	☐
1.10	HM-07	Boom microphone arm: HM-07A temple mount and cheek sleeve, HM-07B 120 mm gooseneck, HM-07C electret capsule and windscreen. The capsule is bare – the preamplifier is on MP-01 at J21, not on the boom	1	BOOM MICROPHONE	no	☐
1.11	HM-08	Quarter-turn battery hatch, three lugs, coin slot, fitted	1 fitted	POD-P1 ENCLOSURE	no	☐
1.12	HM-10	Keyed 18650 cell carrier, fitted behind HM-08. Cannot be inserted reversed (RFQ S-04)	1 fitted	POD-P1 ENCLOSURE	no	☐
1.13	POD-P1	Prototype enclosure base and lid, 163.0 × 143.0 × 62.0 mm closed, Phase 1 only	1	POD-P1 ENCLOSURE	yes	☐
1.14	EEG-CAR-01	Carrier assembly Rev B: 150.0 × 130.0 × 1.60 mm, four layers , 211 reference designators, on the MP-01 plate (146.0 × 126.0 × 3.0 mm), with thirteen module assemblies of twelve types – twelve on MP-01 and the ESP32-S3-DevKitC-1 inserted directly into J6 and J7	1 fitted	POD-P1 ENCLOSURE	yes (unit serial)	☐
1.15	WH-01	Electrode cable, 12-way screened, helmet to J14	1 fitted	HELMET HM-01	no	☐
1.16	WH-02	Light cable, 10-way, helmet to J30	1 fitted	HELMET HM-01	no	☐
1.17	WH-09	USB-B to USB-C panel pigtail, isolator module to the host aperture. Interim, and a live non-conformance against RFQ E-24: the only named isolator module presents USB-B and E-24 asks for USB-C. Fitted until an isolator module with a USB-C host connector is qualified	1 fitted	POD-P1 ENCLOSURE	no	☐
2 ELECTRODES						

#	Part	Description	Qty	Pocket	Serialised	Tick
2.1	—	Ag/AgCl ear-clip references	2	EAR CLIPS + EMG LEADS	no	☐
2.2	—	EMG snap-to-DIN 42802 leads, 1.0 m: red EMG1 cheek, yellow EMG2 chin, green EMG3 neck	3	EAR CLIPS + EMG LEADS	no	☐
2.3	—	Disposable EMG snap pads, pack of 30 (Ambu BlueSensor N or Kendall)	1 pack	CONSUMABLES	no	☐
3 AUDIO						
3.1	—	Closed-back headphones, 32 to 64 Ω per RFQ A-04 as restated; the shipped model is the ATH-M20x at 47 Ω , and the calibrated output level is measured per model into the 47.0 Ω load of TST-EEG-004 T17. 3.5 mm plug, ≥ 1.2 m lead	1	HEADPHONES	no	☐
4 POWER AND CABLES						
4.1	—	Protected 18650 cell ≥ 3000 mAh, UN 38.3, in the HM-10 keyed carrier	1 fitted	POD-P1 ENCLOSURE	lot recorded	☐
4.2	—	Protected 18650, charged spare	0 in circulation	SPARE CELL (empty)	—	☐
4.3	—	USB-C to USB-A cable, 1.0 m, USB 2.0 (A-07)	1	CABLES + CHARGER	no	☐
4.4	—	USB-C to USB-C cable, 1.0 m, USB 2.0 (A-07). One of lines 4.3 and 4.4 is the host lead ; the host connection is a socket at the panel, not a captive cable	1	CABLES + CHARGER	no	☐
4.5	—	5 V 2 A USB charger, EU plug, CE (A-07)	1	CABLES + CHARGER	no	☐
4.6	—	microSD 32 GB high-endurance, seated in the J20 breakout	1 fitted	POD-P1 ENCLOSURE	lot recorded	☐
5 CONSUMABLES (resealable labelled pouch, A-05)						
5.1	—	Conductive EEG paste, 100 g tube	2	CONSUMABLES	no	☐
5.2	—	Abrasive prep gel, 100 g	1	CONSUMABLES	no	☐
5.3	—	Saline wipes	30	CONSUMABLES	no	☐
5.4	—	Cotton buds, pack	1	CONSUMABLES	no	☐
5.5	—	Blunt-tip syringes for the gel ports	4	CONSUMABLES	no	☐

#	Part	Description	Qty	Pocket	Serialised	Tick
6 DOCUMENTS						
6.1	IFU-EEG-014	Laminated A5 quick-start card, artwork QSC-EEG-001: sections 2, 3, 5, 7, 10 and 12 of IFU-EEG-014 condensed to two sides	1	case lid wallet (sections 2.3 and 3.2)	no	☐
6.2	IFU-EEG-014	Laminated A5 placement guide, artwork PLG-EEG-001: IFU-EEG-014 section 13 entire, with the head diagram and the face diagram. Neither diagram has been drawn (IFU-EEG-014 section 16 item 7), so no artwork exists to print and this line cannot yet be ticked	1	case lid wallet, beside the quick-start card	no	☐
6.3	IFU-EEG-014	Full IFU, sections 1 to 13, A5 booklet. IFU-EEG-014 section 14 allows the runner's on-screen help panel instead of the booklet; if the programme takes that option this line is struck by an ECO , so that a packer never decides it	1	case lid wallet	no	☐
6.4	ART-PACK-01	Laminated A5 packing photograph (M-06)	1	case lid wallet	no	☐
6.5	—	Per-unit calibration and test record, printed A4	1	case lid wallet , beside the quick-start card	per serial	☐
6.6	ART-DIS-01	Disinfection guide card (A-05)	1	CONSUMABLES	no	☐
6.7	ART-RET-01	Return-shipping instruction sheet	1	carton label pocket	no	☐
6.8	—	Pre-paid return label	1	carton label pocket	tracking no.	☐
7 MUST NOT BE PACKED – initial each line						
7.1	HM-09	Service key for cup release. Operator only, one per operator	0	—	—	☐
7.2	—	Loose lithium cell of any kind (section 7)	0	—	—	☐
7.3	—	Ultrasonic tooling, spare frames, spare cups beyond the two of line 1.4	0	—	—	☐

#	Part	Description	Qty	Pocket	Serialised	Tick
7.4	—	Headband or cap. Withdrawn as a kit item: the eight electrodes are fixed to the HM-01 frame at manufacture, so a headband with fixed holders at the same eight sites would duplicate them. RFQ A-03 is met by lines 1.7 and 1.8	0	—	—	☐

Footer, signed by two people: *kit complete, case closed, hasps latched, seal number* _____ *applied, outer carton sealed.* Packer _____ Checker _____ Date _____.

The five printed items, and why group 6 was renumbered at this issue. IFU-EEG-014 Rev B section 14 lists five printed items in the case lid pocket: the quick-start card, the placement guide, the full IFU, the per-unit calibration record and the packing photograph. This list carried three of them, and IFU-EEG-014 section 16 item 14 records the shortfall as this document's to close. Lines 6.2 and 6.3 are added for the placement guide and the full IFU, and the rest of group 6 is renumbered so that the five lid-wallet items read together in the order the wallet holds them. Nothing outside this table cites a group 6 line number: the packing photograph of section 2.3 numbers its call-outs to the KPL lines, and it does not exist yet. Two of the five cannot be ticked today and say so on their own line – the placement-guide diagrams are not drawn, and both artwork identifiers QSC-EEG-001 and PLG-EEG-001 are unregistered and collide numerically with RFQ-EEG-001 (IFU-EEG-014 section 16 item 12). A line that cannot be ticked is still the right line to carry: the packer is the person who would otherwise discover the gap at the case, one kit at a time.

Pocket names checked against the cut files. Every pocket named in the column above is one of the nine Rev C bays of section 2.2, and all nine are used: HELMET HM-01, HEADPHONES, POD-P1 ENCLOSURE, BOOM MICROPHONE, SPARE CUPS + KEYLESS SPARES, SPARE CELL, CONSUMABLES, EAR CLIPS + EMG LEADS and CABLES + CHARGER. The two destinations that are not foam bays are deliberate: the case lid wallet, which Rev C has in place of the two deleted card pockets, and the outer carton's label pocket. No line points at a bay of the withdrawn Rev B pair. Read against the seven cut files themselves at this issue –

mech/CASE-00_foam_layer_1.dxf to _7.dxf – layers 1 and 2 carry all nine bay legends, layer 3 carries five (HELMET, HEADPHONES, POD-P1, CONSUMABLES, CABLES + CHARGER), layer 4 two (HELMET, HEADPHONES) and layers 5 to 7 the helmet bay alone, which is exactly the “layers cut” column of the section 2.2 schedule. The patient-contact consumables the kit is useless without are on the list and in a bay: the two Ag/AgCl ear-clip references at line 2.1 and the three EMG DIN leads at line 2.2, both in EAR CLIPS + EMG LEADS, with the disposable snap pads at line 2.3 in CONSUMABLES. What this check does **not** establish is fit: six of the nine bays hold parts that are not dimensioned anywhere in the package, so a name matching a name is not a part matching a pocket, and section 2.2 and open item 1 say which six.

1.2 Corrections made to the kit BOM Rev B

BOM item	As written	Corrected here	Reason
8	microSD breakout, "4-bit SDMMC capable"	one-bit SDMMC	ECO-EEG-009: the three data lines drive the contact-light shift register
14	"112 footprints" on a two-layer 130 × 124 mm carrier	211 reference designators on a 150.0 × 130.0 mm four-layer carrier	design.py; the layer and size change is explained above
16	HM-04 qty 10 per kit	8 bonded, 2 spare bodies held as build stock, not kit content. PARTS-EEG-019 §3.4 reads the same figure as 8 fitted + 2 spare per kit; the two documents disagree and the packer follows this list until PARTS is reconciled	DSN-EEG-002 Rev E section 10, a superseded section – see the note below this table. It is cited for the eight bonded assemblies only; its own quantity column reads HM-04 as 8 + 2 spare <i>per kit</i> , which is PARTS-EEG-019's figure and not this one, so it does not support the spare-body half of this correction. That half is this document's
17	"HM-07 battery hatch"	HM-08 battery hatch; HM-07 is the boom arm	PARTS-EEG-019 §3, part-ID register
17	HM-08 as one line, "hatch and keyed cell carrier"	split: HM-08 is the hatch only (line 1.11); the keyed 18650 carrier is HM-10 (line 1.12)	PARTS-EEG-019 §3.2
18	POD-P1, qty 1 at every phase, 146.8 × 140.8 × 44 mm	Phase 1 only; 163.0 × 143.0 × 58.0 mm external, 158.0 × 138.0 × 55.5 mm internal, 62.0 mm closed with the lid	mech_gen.py; RFQ M-01: Phases 2 and 3 use the HM-01 occipital shell

BOM item	As written	Corrected here	Reason
19	“TPU comfort pads ×4 and chin-cup liner”, qty 1	4 fitted + 4 spare, liner 1 fitted + 1 spare	DSN-EEG-002 Rev E section 10, a superseded section – see the note below this table. Its HM-02 row reads 4 + 4 spare and its HM-06 row calls the liner consumable, and that is all that is taken from it; the liner’s 1 + 1 split is this document’s
20	Headband or cap (RFQ A-03)	Line deleted. A-03 is rewritten to cover the chin strap HM-06 and the occipital yoke HM-03, both already on the frame	The eight electrodes are fixed to HM-01 at manufacture (IFU-EEG-014 §13.1)
24	“20-way FFC to carrier”	two cables: 12-way screened electrode cable to J14, 10-way light cable to J30	ECO-EEG-014
29	Headphones “32 Ω”	32 to 64 Ω ; the ATH-M20x is 47 Ω and the test load is 47.0 Ω	RFQ A-04 as restated; TST-EEG-004 T17
31	Consumables, no cotton buds, no disinfection guide	both added (lines 5.4, 6.3)	RFQ A-05
32	One line, “USB-C to USB-A and USB-C to USB-C, 2 m”	two lines, 1.0 m each	RFQ A-07
33	WH-08 captive host lead through a cable gland	Deleted from the Phase 1 build. The host connection is a socket in a gasketed aperture over the isolator module’s own connector; the captive lead is a Phase 2 item for the helmet shell. WH-09 pigtail added as line 1.17	RFQ E-24 non-conformance, stated at line 1.17

BOM item	As written	Corrected here	Reason
34	Case “~340 × 250 × 210 mm, Nanuk 915 / Peli 1450”	envelope enlarged and those two models withdrawn as too shallow	section 3
35	“DXF supplied for top layer”	both layers supplied, and both superseded by the Rev C stack of section 2.4	section 2
—	No line for the room-microphone module at J28	included in the carrier assembly, line 1.14	ICD-EEG-006 §1
—	No line for MP-01 module plate	included in the carrier assembly, line 1.14	ICD-EEG-006 §1
—	Per-unit calibration record packed in a foam card slot	packed in the case lid wallet beside the quick-start card	TST-EEG-004 §12 and T18
—	No line for the placement guide PLG-EEG-001 or for the full IFU	both added as lines 6.2 and 6.3, and group 6 renumbered so the five lid-wallet items read in order	IFU-EEG-014 Rev B §14 lists five printed items in the lid pocket; §16 item 14 records that this list carried three
—	Charged spare 18650 cut a foam pocket but had no BOM line	quantity is zero in circulation	section 7

On the two citations of DSN-EEG-002 Rev E section 10. That section is the frame parts table, and it is **superseded on three counts and has not been re-issued**: it has no HM-10 line at all, because the keyed cell carrier was still folded into HM-08 when it was written (corrected at BOM item 17 above from PARTS-EEG-019 section 3.2); it describes CASE-00 as the withdrawn two-sheet insert with both sheets supplied as DXF, where section 2 of this document now cuts the seven-layer Rev C stack and the two Rev B files are deleted; and its HM-04 quantity is 8 + 2 spare per kit, which is the figure BOM item 16 above departs from. RISK-EEG-011 Rev B section 6.1 records that DSN-EEG-002 Rev E must be corrected, for its section 7 statement that a charged spare cell travels in the kit, and no re-issue has been made. The two rows above therefore cite it as the origin of a number, not as a live source, and where this document and that section disagree, this document governs the packing. A reader wanting the current frame parts list takes it from PARTS-EEG-019 Rev B.

Verification: the programme’s goods-in in Brussels unpacks two kits per batch against this list (RFQ 9.3, QP-EEG-010 A-21). Sign-off: programme QA.

2. Foam insert CASE-00

2.1 Material and construction

Property	Value	Verification
Material	Closed-cell polyethylene (PE), not PU, not XLPE	Supplier certificate per lot
Density	28 to 33 kg/m ³ to ISO 845	Supplier certificate
Compression stress at 25 %	30 to 50 kPa to ISO 3386-1	Supplier certificate
Colour	Black through-dyed	Visual
Layers, Rev B	Two, loose-stacked, not bonded	Visual
Layers, Rev C	Seven loose-laid 25 mm layers, not two. Section 2.4 explains why two 25 mm sheets cannot hold this kit	First article against the section 2.2 Rev C schedule
Stack depth, Rev C	175.0 mm , seven layers of 25 mm	Caliper, 4 points
Thickness as drawn, Rev B	25 mm each layer, per the title text in both DXFs	Caliper, 4 points per sheet
Sheet size, Rev B	340.0 × 250.0 mm, both layers	Caliper, ±2.0 mm
Sheet size, Rev C	516.0 × 390.0 mm : the Peli 1560's published 518 × 392 mm internal footprint minus 2 mm on each axis, settled in section 3.2	Caliper against the measured shell before any foam is cut . The seven cut files are drawn to the published figure and are held: a shell that measures short does not make them tight, it makes them scrap
Pocket tolerance	±1.0 mm	First article, then AQL per QP-EEG-010
Lid foam	15 mm PE or PU convoluted, plain, no pockets, over the helmet bay only	Visual

Sign-off on the material: programme QA against the supplier certificate at IQC.

2.2 The foam pocket schedule and the cut files

This section is the only authoritative foam pocket schedule in the package. SVC-EEG-013 Rev B section 2 R12, IFU-EEG-014 Rev B section 1 and PARTS-EEG-019 Rev B section 2.3 cite this section and do not restate the pockets. Two schedules appear below and they are not interchangeable. The **Rev B** schedule is what the two withdrawn DXF files contained; it is recorded because they were released in package

v2.1 and a manufacturer may still hold them, and it cannot pack this kit for the reasons in section 2.4. The **Rev C** schedule is what will be cut, and at this issue it is also what the package supplies: `tools/mech_gen.py foam_dxf()` writes the seven Rev C layer files listed below it and cannot emit the Rev B pair any more. Anyone copying pocket sizes takes them from the Rev C table.

Every cut file is supplied 1:1, `$INSUNITS = 4` (millimetres), datum at the bottom-left corner of the sheet with X to the right and Y up. Two named layers are used and they mean different things: **CUT** carries the sheet outline, every bay profile and the $\varnothing 22$ mm finger reliefs; **TEXT** carries the bay legends and the title block and is **not an output layer** – nothing on layer TEXT is cut or engraved. v1 shipped one file with the legends on the same layer as the cuts, which a laser would have cut out of the foam. That is fixed.

The two Rev B files are **withdrawn, and deleted from mech/ at this issue** rather than left beside the Rev C stack, because a superseded cut file next to a live one is eventually cut. What they contained is recorded here and nowhere else.

CASE-00_foam_top_layer.dxf – top layer, as cut (Rev B, deleted):

Legend	Origin (x, y)	Size (mm)	Finger relief
HELMET HM-01	12, 60	200 × 176	$\varnothing 22$ at (112, 72)
HEADPHONES	220, 130	108 × 106	$\varnothing 22$ at (274, 142)
CONSUMABLES	220, 60	108 × 62	$\varnothing 22$ at (274, 72)
CASE LID CARD	12, 6	316 × 46	$\varnothing 22$ at (170, 18)

CASE-00_foam_bottom_layer.dxf – bottom layer, as cut (Rev B, deleted):

Legend	Origin (x, y)	Size (mm)	Finger relief
POD-P1 ENCLOSURE	14, 150	152 × 90	$\varnothing 22$ at (90, 162)
EAR CLIPS + EMG LEADS	176, 150	150 × 90	$\varnothing 22$ at (251, 162)
CABLES + CHARGER	14, 84	152 × 58	$\varnothing 22$ at (90, 96)
SPARE CELL	176, 84	70 × 58	$\varnothing 22$ at (211, 96)
SPARE CUPS + KEYLESS SPARES	252, 84	74 × 58	$\varnothing 22$ at (289, 96)
BOOM MICROPHONE	14, 14	152 × 62	$\varnothing 22$ at (90, 26)
QUICK-START CARD	176, 14	150 × 62	$\varnothing 22$ at (251, 26)

All Rev B pockets are through-cuts in their own 25 mm layer. **Rev C deletes the two card pockets** – CASE LID CARD and QUICK-START CARD – because all five printed items travel in the case lid wallet: the quick-start card, the placement guide, the full IFU,

the calibration certificate and the packing photograph (section 1.1 lines 6.1 to 6.5, IFU-EEG-014 Rev B section 14). That closes the mismatch IFU-EEG-014 section 14 records and cannot fix from its own side.

CASE-00 Rev C – the schedule that will be cut. Sheet 516.0 × 390.0 mm, seven loose-laid PE layers of 25 mm, 175 mm of stack, layer 1 uppermost, datum at the bottom-left corner with X to the right and Y up. Origin and size are the bay opening in plan. “Layers cut” says which layers the opening passes through; the first uncut layer beneath is the floor of the bay. “Packer” is a plain uncut PE pad, cut from offcut, laid in the bottom of the bay so the part sits near flush in an opening whose depth can only be a multiple of 25 mm. Finger reliefs are ø22 mm and are cut through the same layers as their bay.

Legend	Origin (x, y)	Size (mm)	Layers cut	Cut depth	Packer	Finger relief
HELMET HM-01	14, 14	197 × 236 in layers 2 to 7; 181 × 220 at (22, 22) in layer 1 , which is the 8 mm shelf the halo lands on	1 to 7, open to the case floor	175	none	none – the helmet is lifted by the halo
HEADPHONES	219, 14	181 × 161	1 to 4	100	17	ø22 at (309, 28)
POD-P1 ENCLOSURE	219, 183	169 × 149	1 to 3	75	10	ø22 at (303, 197)
BOOM MICROPHONE	14, 258	197 × 58	1 to 2	50	at the trial pack	ø22 at (112, 272)
SPARE CUPS + KEYLESS SPARES	14, 324	96 × 52	1 to 2	50	at the trial pack	ø22 at (62, 338)
SPARE CELL	118, 324	93 × 52	1 to 2	50	at the trial pack	ø22 at (164, 338)
CONSUMABLES	408, 14	94 × 140	1 to 3	75	at the trial pack	ø22 at (455, 28)
EAR CLIPS + EMG LEADS	408, 162	94 × 100	1 to 2	50	at the trial pack	ø22 at (455, 176)
CABLES + CHARGER	408, 270	94 × 106	1 to 3	75	at the trial pack	ø22 at (455, 284)

Nine bays, not eleven: the two card pockets are gone and all five printed items live in the lid wallet. Every bay is at least 3 mm larger than its part on every axis it is dimensioned against, every web between bays is at least 8 mm, and every bay is at least 14 mm from the sheet edge. The layout closes: 14 + 197 + 8 + 181 + 8 + 94 + 14 = 516 mm across, and 14 + 236 + 8 + 58 + 8 + 52 + 14 = 390 mm up. The 181 × 36 mm patch above the POD-P1 bay is left as solid foam.

The depth column is the half of this schedule Rev B did not have. HEADPHONES is cut through four layers, 100 mm, with a 17 mm packer, so an 80 mm folded headphone sits in an 83 mm well. POD-P1 is cut through three layers, 75 mm, with a 10 mm packer, so the 62 mm closed pod sits in a 65 mm well. HELMET HM-01 is cut through all seven and is open to the case floor, because the helmet is 158 mm tall and hangs from the shelf rather than sitting on a pocket floor.

Six of the nine bays – BOOM MICROPHONE, SPARE CUPS + KEYLESS SPARES, SPARE CELL, CONSUMABLES, EAR CLIPS + EMG LEADS and CABLES + CHARGER – hold parts that are not dimensioned anywhere in package v2. Their plan

sizes above are what the layout leaves once the three dimensioned bays are placed, and their layer counts are a first estimate. What can be said is that each of the six has **more volume than the Rev B pocket of the same name**, between 1.4 and 5.9 times as much, because the extra depth more than pays for a narrower plan: the ear-clip bay, the only one that loses plan area, goes from 338 cm³ to 470 cm³. They are confirmed or corrected at the trial pack of section 2.4, and their packers are cut then. Saying a bay is provisional is not the same as saying it is wrong; it means nobody has yet held the part against it.

The Rev C cut files. Seven files, one per layer, written by `tools/mech_gen.py` `foam_dxf()` from the table above and from nothing else. A bay is drawn in a file only where the schedule cuts it through that layer, so the seven are not interchangeable and each one names itself, its layer number and its bay count in its own TEXT layer.

File	Part id	What is cut in this layer
mech/ CASE-00_foam_layer_1.d xf	CASE-00-01	all nine bays, with the helmet bay at the 181 × 220 mm shelf opening
mech/ CASE-00_foam_layer_2.d xf	CASE-00-02	all nine bays, helmet bay at the full 197 × 236 mm
mech/ CASE-00_foam_layer_3.d xf	CASE-00-03	HELMET HM-01, HEADPHONES, POD-P1 ENCLOSURE, CONSUMABLES, CABLES + CHARGER
mech/ CASE-00_foam_layer_4.d xf	CASE-00-04	HELMET HM-01, HEADPHONES
mech/ CASE-00_foam_layer_5.d xf	CASE-00-05	HELMET HM-01
mech/ CASE-00_foam_layer_6.d xf	CASE-00-06	HELMET HM-01
mech/ CASE-00_foam_layer_7.d xf	CASE-00-07	HELMET HM-01

Layers 5, 6 and 7 are the same cut, and they are supplied as three files anyway so that a sheet cannot be laid into the stack at the wrong height with nothing on it saying so. The identifiers CASE-00-01 and CASE-00-02 are re-used from the two deleted Rev B files at Rev C; CASE-00-03 to CASE-00-07 are new, and PARTS-EEG-019 Rev B carries rows for the first two only and has still to register the other five.

What the Rev C table does not specify, and what the cut files do with it. The **corner radii**: the table gives none, so the files carry the radii of the deleted Rev B file, which took them from package v1 – R40 on the helmet bay because the frame is domed, R10 on the two large equipment bays and on the ear-clip bay, R8 on the rest, R14 on the

sheet. Section 2.3 orders a steel-rule die at Phase 3 and steel rule cannot be folded to a true 90 degree inside corner, so a profile with no stated radius becomes whatever radius the die maker chooses. The **packers**: the table gives a thickness for two of the nine bays and “at the trial pack” for the other six, and no packer outline anywhere, so no packer is drawn in any file and all of them are cut from offcut at the trial pack. The **title block**: its wording and its position, in the solid patch above the POD-P1 bay and in the bottom edge margin, are this document’s and not the table’s.

And one thing the schedule has wrong, which the cut files reproduce rather than quietly correct. Every finger-relief centre in the Rev C table sits at least 3 mm inside its own bay opening, so each $\varnothing 22$ mm circle falls entirely within foam that the bay profile already removes, and it cuts nothing. A relief only does work when it straddles the bay wall, and that cannot be had by moving the circle outwards: a straddling $\varnothing 22$ mm relief needs 11 mm of foam beyond the wall and every web in this layout is 8 mm. The two ways out are a scallop cut in the first uncut layer beneath the bay, which is where a finger actually goes in a deep well, or a re-columned sheet with wider webs. Both are a Rev D of this schedule and neither is invented here. The reliefs are cut where the table puts them, they cost nothing, and open item 12 of section 9 carries the defect.

2.3 Cutting: laser or die

Quantity	Method	Kerf compensation	Why
Phase 1, 2 kits	Pluck foam, hand-fitted (kit BOM item 35 alternate)	n/a	The layout is not yet verified against a printed helmet
Phase 2, 10 kits	Laser, CO ₂ , sealed edge	Profiles drawn on size; the cutter offsets 0.5 mm per side	Seventy sheets – ten kits of seven layers – does not repay five dies
Phase 3, 10 to 40 further kits, 25 to 50 in total	Steel-rule die, one die per distinct layer profile: five, not one. Layers 1, 2, 3 and 4 are four different profiles and layers 5, 6 and 7 are the fifth, all at one 25 mm thickness	None; profiles are the die line	Tooling repays itself past about 30 sheets per profile, and the layer 5 to 7 die cuts three sheets per kit

PE cuts by melting. The edge seals, which is wanted, but PE **cannot be raster-engraved legibly** – the letters close up as they cool. The legends of RFQ M-06 are therefore **printed adhesive tags**, artwork ART-LBL-04, 60 × 12 mm, white on black matt polyester with permanent acrylic adhesive, one per bay. On a shallow bay the tag is bonded to the pocket’s rear wall at mid-depth; on the helmet, pod and headphone bays, which are now deep wells, it is bonded to the **top face of the foam beside the opening**, because a tag 80 mm down a well cannot be read. Cap height 8 mm. Legend text is exactly the schedule legend string, so the tag and the cut file cannot drift, and `tools/artwork_gen.py` now enforces that rather than asserting it: the tags are generated by reading the section 2.2 Rev C schedule, so a legend cannot be re-typed differently on a tag than it is cut in a file.

The 8 mm cap height does not fit the 60 mm tag, and none of the nine legends fits.

This is a measurement, not an opinion. Set on one line at an 8 mm cap height in the face the artwork uses, HEADPHONES – the least demanding of the nine – needs about 88 mm of tag, CONSUMABLES 96 mm, POD-P1 ENCLOSURE 126 mm, EAR CLIPS + EMG LEADS 154 mm, SPARE CUPS + KEYLESS SPARES 200 mm, and the section 7 form of the spare-cell tag 320 mm. A narrower face buys perhaps fifteen per cent, not the factor of one and a half the least demanding needs and not the factor of five the worst needs. The generated tags are therefore drawn at the 60 x 12 mm section 4.2 fixes, at the largest cap height that fits on at most two lines, which runs from 5.36 mm on HEADPHONES down to 2.68 mm on the spare-cell tag. **Allowed two lines, a tag of about 171 x 21 mm carries every legend at 8 mm**, the spare-cell tag again setting the width. That is a change to section 4.2's tag size and to this paragraph's cap height, and it is open item 15 of section 9 rather than a change made here. Verification: every legend present and legible after the kit has been packed and unpacked once (QP-EEG-010 FK-04). Sign-off: QA.

Five printed items are fixed inside the case lid, not in the foam, in an A4 document wallet retained by the lid: the laminated A5 quick-start card (A-06), the laminated A5 placement guide, the full IFU, the printed A4 calibration certificate and the laminated A5 packing photograph (M-06). That is the list of IFU-EEG-014 Rev B section 14, and it is now also lines 6.1 to 6.5 of section 1.1. *Was: three items – photograph, card and certificate* – which is the count IFU-EEG-014 section 16 item 14 raised against this document, because the two guides it left out have nowhere else to travel: Rev C cuts no paper bay at all.

The photograph is taken of the first Phase 1 kit, 300 dpi, A5 landscape, call-outs numbered to the KPL line numbers. Until a kit exists, a rendered layout marked **PROVISIONAL – REPLACE WITH PHOTOGRAPH AT FIRST BUILD** is used. **That render now exists:** graphics/labels/ART-

PACK-01_packing_layout_provisional.svg, with its control sheet ART-
PACK-01_packing_layout_provisional.pdf. It draws the nine Rev C bays at true relative scale on the 516.0 x 390.0 mm sheet and numbers each one with every packing-list line whose pocket column names it, both read out of this document at generation time. It is a plan of layer 1 and it shows no depth, so it says where a part goes and not that it fits; open item 1 still holds. Open item 7 is closed to the extent that the stand-in exists and is marked as one; the photograph itself is still owed at the first build.

2.4 What is not yet right, stated plainly

CASE-00 Rev B is a coherent cut file and it cannot pack this kit. No kit is packed to CASE-00 Rev B, and at this issue its two files are deleted and replaced by the seven Rev C files of section 2.2. There are two independent failures and the second is the worse of the two.

Footprint. Five of the eleven Rev B pockets cannot accept the item their legend names, measured against the part geometry in mech_gen.py and DSN-EEG-002 Rev E.

Depth. Every Rev B pocket is a through-cut in a 25 mm sheet, so no pocket can hold anything taller than 25 mm, and the three largest items in the kit are 62, 80 and 158 mm tall. This was missed in Rev A of this document, which checked X and Y only.

Pocket	As cut	Item and its real size	Shortfall in X / Y	Shortfall in depth
HELMET HM-01	200 × 176 through 25 mm	HM-01 standing on the halo, 191.1 × 229.6 × 158	53.6 mm in Y	133 mm
POD-P1 ENCLOSURE	152 × 90 through 25 mm	POD-P1 closed, 163.0 × 143.0 × 62.0	11.0 mm in X, 53.0 mm in Y	37 mm
HEADPHONE S	108 × 106 through 25 mm	ATH-M20x folded, about 175 × 155 × 80	about 67 mm in X, 49 mm in Y	55 mm
CASE LID CARD	316 × 46 through 25 mm	A5 laminated card, 210 × 148	102 mm in Y	none; pocket deleted
QUICK- START CARD	150 × 62 through 25 mm	A5 laminated card, 210 × 148	60 mm in X, 86 mm in Y	none; pocket deleted

The remaining Rev B pockets – EAR CLIPS + EMG LEADS, CABLES + CHARGER, CONSUMABLES, BOOM MICROPHONE, SPARE CUPS + KEYLESS SPARES and SPARE CELL – **cannot be checked at all**, because the coiled electrode set, the consumables pouch, the charger body and the boom arm are not dimensioned anywhere in package v2. Rev C dimensions them at the trial pack. Not being able to check a pocket is not the same as the pocket being right.

How the helmet is actually held. The helmet cannot be held by a pocket. HM-01 stands 158 mm tall, travels assembled and upright, and must be supported **at the halo** with nothing bearing on the electrode cups, which is the point of the whole v2 helmet design. Two 25 mm sheets give 50 mm of foam in total, so even stacking both layers is 108 mm short. CASE-00 Rev C therefore stops being a two-sheet insert and becomes a **stack of seven loose-laid 25 mm PE layers, 175 mm deep**, cut to the Rev C schedule of section 2.2 so that:

- the helmet bay is a **well through the whole stack**, open to the case floor, with an opening of at least 197 × 236 mm in the layers below the top one;
- the **top layer of the helmet bay is cut about 8 mm smaller all round** than the layers beneath it, forming a shelf. The halo rim of HM-01 lands on that shelf and the cups hang free in the void beneath, touching nothing. The shelf is the only load path;
- the helmet therefore stands proud of the foam by however much of it sits above the halo rim. That figure comes off the first printed HM-01, not off a drawing. The 15 mm convoluted lid foam closes on the halo and provides the retaining force;
- the pod, headphone, electrode, boom, cable, charger, consumable and spare bays are cut through as many layers as their item is deep, with the first uncut layer beneath forming the floor of each and a plain PE packer making up the difference between the item and the 25 mm granularity of the stack. The pod bay is 169 × 149 mm and 65 mm deep over a 10 mm packer; the headphone bay 181 × 161 mm and 83 mm deep over a 17 mm packer;

- every bay is at least 3 mm larger than its part on every axis, with 8 mm of foam between bays and 14 mm to the sheet edge.

The depth budget, in one line. 175 mm of foam stack, plus however much of HM-01 stands above the shelf, plus 15 mm of convoluted lid foam, must fit inside the 229 mm of internal depth the settled shell of section 3.2 publishes. That leaves 39 mm for the helmet to stand proud before the lid foam is compressed at all, and convoluted foam compresses by about half its thickness, so the working allowance is about 45 mm. Whether the halo rim of HM-01 sits within 45 mm of the top of the frame is measured off the first printed frame and is the single measurement that decides whether this stack closes.

The Rev B sheet footprint is dead too, and not by a small margin. The helmet bay alone needs 197×236 mm. On a 340×250 mm sheet that leaves a strip 143 mm wide, and the pod bay needs 169 mm of it. CASE-00 Rev B cannot hold this kit at any depth, in any arrangement. The nine Rev C bays add up to $154,592 \text{ mm}^2$ of opening before a single web or edge margin is drawn, which is more than the whole 340×250 mm sheet. Laid out on the rules above they occupy **516×390 mm**, and that is the figure the case requirement of section 3.2 is written from. The “about 420×350 mm” carried by the first issue of this revision was an estimate rather than a layout, it was smaller than the bay openings it was meant to hold, and it is withdrawn.

CN-PKG-02, entered in the ECO-EEG-016 register: CASE-00 goes to Rev C before any kit is packed. The Rev C schedule is dimensioned in section 2.2 and no longer waits on the case decision, because section 3.2 settles the case, and at this issue the seven Rev C cut files are drawn to it. What still waits is the **cut**: the sheet is drawn to a case internal dimension nobody has measured, six of the nine bays hold parts that are not dimensioned anywhere in the package, and every packer is cut at the trial pack. Rev C is confirmed against a physical trial pack with the first printed HM-01 and the first bought shell, not against a drawing, because nothing here has been printed or bought. It must satisfy, in order: the helmet stands assembled and upright on the halo with nothing bearing on the cups (DSN-EEG-002 Rev E section 11); the foam stack plus the 158 mm helmet plus the lid foam is inside the case internal depth, on the budget above; and every bay meets the clearance rule above. The two card slots are deleted and the cards live in the lid. The parts decide the case, not the other way round. Owner: programme mechanical. Gate: the trial pack is a Phase 1 acceptance item, signed by the programme before Phase 2 is released.

3. Travel case

3.1 The sizes in the earlier documents, reconciled

RFQ-EEG-001 Rev C M-05 said internal approximately $300 \times 220 \times 110$ mm, Nanuk 910 / Peli 1400 class. Kit BOM Rev B item 34 said approximately $340 \times 250 \times 210$ mm, Nanuk 915 / Peli 1450, which agrees with the Rev B foam sheet. RFQ-EEG-001 Rev E M-05 carries the larger of the two. AVL-EEG-017 Rev C K21 restates it no longer: at this issue K21 cites section 3.2 for the envelope, names the same shell this section names and states the same measurement rule, so the fourth figure that was in circulation – K21’s “internal $\geq 340 \times 250 \times 210$ mm (M-05)” – is withdrawn. **The three that remain are all too small**, and section 2.4 is why: the bay schedule needs 516×390 mm of internal footprint and at least 185 mm of internal depth in the base.

The helmet decides it. HM-01 is $191.1 \times 229.6 \times 158$ mm and travels **assembled and upright, supported at the halo**: the participant never assembles an instrument at home. A 110 mm deep case cannot take a 158 mm tall helmet in any orientation that keeps the cups off the foam, and a 340×250 mm case cannot take the helmet bay and the pod bay side by side.

CN-PKG-03, entered in the ECO-EEG-016 register: RFQ-EEG-001 M-05's "internal size approximately $340 \times 250 \times 210$ mm" is superseded by the requirement in section 3.2 and must be corrected at the next RFQ revision, and AVL-EEG-017 K21 must cite section 3.2 rather than restate M-05. Until those revisions are issued, **M-05 as written and this document disagree, and this document governs on the geometry** because the bay schedule is derived from part dimensions and M-05's figure is not.

The named example models are a second problem. **Nanuk 915 and Peli 1450 are about 155 mm deep internally**, so the kit BOM's own example models fail its own dimension. Both are withdrawn.

3.2 The internal envelope, settled

This section is the single home of the travel-case internal dimensions. AVL-EEG-017 Rev C K21 and PARTS-EEG-019 Rev B CASE-01 cite it; RFQ-EEG-001 M-05 is corrected to it at the next revision under CN-PKG-03. One shell, one transcription, one set of numbers, so that the foam file of section 2.2 and the carton of section 6 can both be drawn from the same figures instead of from two.

Two transcriptions of the same shell were in circulation. Earlier issues of this document quoted the Peli 1560 as approximately $508 \times 330 \times 198$ mm internal; AVL-EEG-017 K21 quotes **$518 \times 392 \times 229$ mm**. Both are the manufacturer's published figures copied by hand and the difference is not resolvable from paper. **AVL-EEG-017 K21's figures are adopted and this document's $508 \times 330 \times 198$ mm is withdrawn as a mis-transcription**, so that the package carries one number for one shell. They remain published-and-unverified: the first shell received is measured, in the base and in the lid separately, before a single sheet of foam is cut.

Requirement	Value	Verification	Sign-off
Usable internal footprint, lid closed	$\geq 516 \times 390$ mm, derived bay by bay from the CASE-00 Rev C schedule of section 2.2. Larger than RFQ M-05, see CN-PKG-03	Measure the first shell received, before any foam is cut. The seven cut files are drawn to the published figure and are not released for cutting until that measurement is in	Programme mechanical
Usable internal depth	≥ 210 mm in total, of which ≥ 185 mm in the base, so the 175 mm foam stack stands in the base and the helmet stands proud into the lid volume on the depth budget of section 2.4	Measure the base and the lid separately at goods-in	Programme mechanical

Requirement	Value	Verification	Sign-off
Ingress	IP67, watertight and dust-tight, gasketed lid (M-05)	Supplier datasheet filed at IQC; 30 min immersion at 1 m on one Phase 1 shell	QA
Pressure equalisation	Automatic valve, hydrophobic membrane, so the case opens after air freight	Present and unobstructed; no label may cover it	Packer, at every despatch
Latches	At least two lockable hasps accepting a 5 mm shackle	Visual, and one open/close cycle at goods-in	Packer
Handle	Integral moulded, no wheels – the case must sit inside a carton. The settled shell is a large-format case offered in wheeled and unwheeled variants; the purchase order must name the unwheeled variant	Visual	QA
Temperature	–20 to +50 °C operating	Supplier datasheet	QA
Lid	Retains the A4 document wallet holding all five printed items – quick-start card, placement guide, full IFU, calibration certificate and packing photograph (section 1.1 lines 6.1 to 6.5) – without foam. The wallet is the only place paper travels: Rev C cuts no card bay	Trial pack, with the five items in the wallet and the lid closed	QA

Requirement	Value	Verification	Sign-off
Empty mass	The ≤ 4.0 kg target of Rev A is not met and is withdrawn. No shell that meets the envelope above is a 4.0 kg case; the settled shell's published empty mass without foam is about 5.1 kg. Chargeable weight is set by carton volume, not by mass (section 6), so the target bought nothing it claimed to buy. The requirement that replaces it: the lightest shell that meets the envelope, with its measured empty mass recorded	Weigh the first shell	Programme

The shell. The **Peli 1560** is the baseline, at a published internal $518 \times 392 \times 229$ mm. It meets the footprint with 2 mm to spare on each plan axis and the total depth with 19 mm to spare. Two things about it are measured rather than read. Peli publishes one interior depth and does not split base from lid, so the ≥ 185 mm base figure is the one number that can still sink this shell. And 2 mm of plan margin is inside the tolerance of a moulded case, so a shell that measures short does not make the cut file tight, it makes it scrap. Both are why the shell is measured before the foam is ordered and not after. The seven Rev C cut files of section 2.2 exist and they are **not released for cutting**: their sheet is 516.0×390.0 mm because the published internal footprint is 518×392 mm, and the sheet is re-drawn to the **measured** footprint minus 2 mm on each axis before a sheet of foam is bought or cut.

AVL-EEG-017 Rev C K21 agrees with this section at this issue. It names the Peli 1560, unwheeled, as the baseline; it quotes the same published $518 \times 392 \times 229$ mm; it carries the Nanuk 960 as the alternate the CASE-00 Rev C layout does not fit; and it states the measure-before-you-cut rule rather than a fourth set of numbers. Where the two ever differ, this section governs and K21 is corrected to it, because the bay schedule is derived from part dimensions and a vendor list is a transcription.

The alternate. AVL-EEG-017 K21 names a Nanuk **960** at a published internal $552 \times 358 \times 226$ mm. It clears the depth, but it is 32 mm short of the 390 mm second axis, so **the CASE-00 Rev C bay layout does not fit it as drawn**. It stays the alternate shell, and choosing it means re-columning the boom, spare-cup, spare-cell and cable bays onto a 550×356 mm sheet, which is a Rev D of CASE-00 and is not drawn. The Nanuk 945 and the Peli 1600 named in earlier issues of this section are withdrawn: neither appears in AVL-EEG-017, and the package carries one approved vendor list.

That leaves the programme on one shell with no drawn alternate, which is recorded as open item 10 in section 9. The model is frozen at the trial pack of section 2.4, and the chosen shell's datasheet and its measured internal dimensions are filed in the device history record. Whichever is chosen, CASE-00 is sheeted to the **measured** internal footprint minus 2 mm on each axis, not to the published one.

The case is part of the BOM, not packaging (M-05). It is never the shipping container (M-07, section 6).

4. Labelling

4.1 Unit label – ART-LBL-01 (RFQ M-03, E-21, TST-EEG-004 T18)

Artwork: graphics/labels/ART-LBL-01_unit_label.svg (the printer's file, at trim size, with the per-unit fields drawn as <<PLACEHOLDER>>), ART-LBL-01_unit_label_specimen.svg (the same label with a real Data Matrix, drawn for TIOV-B-0000) and ART-LBL-01_unit_label.pdf (the control sheet a person signs). All three are generated by tools/artwork_gen.py from the table below.

Property	Value
Size	50 × 25 mm
Substrate	Matt white polyester, 50 µm
Print	Thermal transfer, resin ribbon, black
Finish	Clear over-laminate, 25 µm
Adhesive	Permanent acrylic, service –40 to +90 °C
Placement, Phase 1	POD-P1 lid outer face, on the flat 55 × 30 mm keep-out, clear of the gasket line
Placement, Phases 2 and 3	HM-01 occipital shell rear face
Duplicate	One identical label inside the case lid
Durability	Legible, adherent and Data-Matrix-readable after 50 wipes with 70 % IPA and after 20 simulated round trips

Content, in this order:

1. TIOV-B-0001 – unit serial, section 5, 10 pt minimum
2. HW EEG-CAR-01-B – hardware revision. The revision letter here and the letter in the serial are the same letter and must agree
3. FP 9F2C 4108 BB37 1D0A – ATECC608B public-key fingerprint, 16 uppercase hex in four groups (the example is illustrative)
4. ECC200 Data Matrix, 12 × 12 mm, quiet zone 2 modules
5. RESEARCH INSTRUMENT -- NOT A MEDICAL DEVICE
6. Do not wear while charging (RFQ S-01)
7. Contains Li-ion cell -- do not incinerate
8. one.witysk.org · CC BY-SA 4.0

Data Matrix content, pipe-delimited, no spaces: TIOV-B-0001|EEG-CAR-01-B|<18 hex ATECC factory serial>|<16 hex fingerprint>

The fingerprint is defined once, in **FW-EEG-001 Rev D section 7**, and this document does not restate the derivation. ASM-EEG-007 Rev B section 6.2, QP-EEG-010 Rev B,

TST-EEG-004 Rev C T6 and REG-EEG-012 Rev B cite the same definition, and all of them must return the same string for the same unit. The **serial** is the other way round: FW-EEG-001 section 7 does not define it, section 5 of this document does.

Verification: at T18 the operator scans the Data Matrix and compares it field by field with the provisioning record and with the USB iSerialNumber read at T5b. A mismatch quarantines both the unit and the label. Barcode quality ISO/IEC 15415 grade C or better after the IPA test. Sign-off: production QA, countersigned by the programme on the two kits sampled per batch (QP-EEG-010 A-13).

4.2 Case, carton and cable labels

Ref	Label	Size	Substrate	Placement
ART-LBL-02	Kit ID plate: KIT-007 in 24 pt, Code-128 of the kit id, unit serial beneath, "TI One Voice research kit – property of TI One Voice vzw, Brussels"	80 × 40 mm	Outdoor white polyester, permanent	Case lid exterior, upper left
ART-LBL-03	Return address and if-found, EN/FR/NL, Brussels address, programme telephone, "Contains no personal data"	100 × 60 mm	Outdoor white polyester, permanent	Case base exterior
ART-LBL-04	Foam bay tags, one per bay, legend text identical to the section 2.2 schedule	60 × 12 mm	Matt polyester, permanent	Bay rear wall at mid-depth, or the top face beside a deep well (section 2.3)
ART-LBL-05	Carton marking set: FRAGILE – RESEARCH INSTRUMENT, THIS WAY UP, gross mass, kit id	printed on the carton	—	Two opposing long faces
ART-LBL-06	Lithium battery mark, UN 3481, telephone number	120 × 110 mm, reduced 105 × 74 mm where the face is too small	White polyester	Outer carton, one side face
ART-LBL-07	Numbered tamper seal	90 × 20 mm	Void-evident, removable	Across each case hasp

No label may cover the pressure-equalisation valve. Sign-off on placement: packer, checked by QA against the placement drawing DRW-LBL-PLACEMENT, which is graphics/labels/DRW-LBL-PLACEMENT_label_placement.svg and its A3 control sheet DRW-LBL-PLACEMENT_label_placement.pdf. It carries four views – the POD-P1 lid, the case lid, the case base and one long face of the outer carton – with every label drawn at its size from this table and dimensioned from the two nearest edges. **The faces and the sizes are this section's and are fixed; the positions on those faces are the drawing's and are a proposal**, because no shell has been measured (section 9 open item 2) and the carton is sized from a published figure (section 6). Dimensioning from the edges is what lets a measured shell change the numbers without changing the arrangement. The valve is a written constraint on that sheet and not a drawn keep-out, because its position on the bought shell is not known.

The SPARE CELL bay tag is specified twice and the two do not agree. This table says the ART-LBL-04 legends are “identical to the section 2.2 schedule”, which makes the tag read SPARE CELL; section 7 says the tag reads SPARE CELL - - DEPOT ONLY, EMPTY IN CIRCULATION. The generated tag carries the section 7 string, because it is the one that tells a packer why the bay is empty, and the collision is open item 16 of section 9.

Cables carry printed heat-shrink sleeves at both ends, 25 mm long, white on black, applied before the connector is terminated (WH-EEG-008):

ID	Cable	Sleeve text
CBL-01	Electrode cable, 12-way screened, helmet to J14	CBL-01 ELECTRODE J14 + unit serial
CBL-02	Light cable, 10-way, helmet to J30	CBL-02 LIGHTS J30 + unit serial
CBL-03	EMG lead 1, red	EMG1 CHEEK
CBL-04	EMG lead 2, yellow	EMG2 CHIN
CBL-05	EMG lead 3, green	EMG3 NECK
CBL-06	USB-C to USB-A, 1.0 m	CBL-06 USB-C / USB-A 1.0 m
CBL-07	USB-C to USB-C, 1.0 m	CBL-07 USB-C / USB-C 1.0 m
CBL-08	Boom microphone pigtail to J18	CBL-08 BOOM J18

The two USB cables are not function-coded, because either can serve either port. The **ports** carry the legends instead, on the pod panel: DATA (isolated) at the host aperture and CHARGE ONLY at J24. The host aperture is a gasketed opening over the isolator module's own connector, not a gland and not a captive lead. On the only named isolator module that connector is USB-B where RFQ E-24 asks for USB-C, so the WH-09 pigtail of packing-list line 1.17 presents USB-C at the panel; that is an interim answer to a live non-conformance, not a settled design. The EMG colour code is carried through from the lead sleeve to the panel legend beside J15/J16/J17 so a participant matches by colour alone.

4.3 The artwork files

ECO-EEG-016 Rev C section 1 registers ART-LBL-01 to ART-LBL-07, ART-PACK-01, ART-DIS-01, ART-RET-01 and DRW-LBL-PLACEMENT as “artwork files controlled as generated artifacts under graphics/” owned by this document. Until 2 September 2026 none of them existed: the register named eleven files, section 1.1 ticked against three of them, section 4.2 made the QA placement check a check against a drawing that was not in the package, and the nine bay tags could not be ordered, only re-typed. They are now generated, into graphics/labels/.

They carry **no revision letter of their own**, only Rev B of this document, which is what ECO-EEG-016 requires. Two files per identifier: a .svg at exact trim size with all text converted to outlines, which is the file a printer is given and which has no font to substitute, and a .pdf control sheet with the title block, the artwork at 1:1 inside trim marks, the variable-data table and the notes, which is the file a person reads and signs. graphics/labels/README_artwork.txt lists every file with its SHA-256.

Identifier	File stem in graphics/labels/	Source of its content
ART-LBL-01	ART-LBL-01_unit_label, and _unit_label_specimen	Section 4.1
ART-LBL-02	ART-LBL-02_kit_id_plate, and _kit_id_plate_specimen	Section 4.2
ART-LBL-03	ART-LBL-03_return_and_if_fo und	Section 4.2
ART-LBL-04	ART-LBL-04_bay_tag_<legend>, nine files	Section 2.2 Rev C schedule, and section 7 for the spare-cell tag
ART-LBL-05	ART-LBL-05_carton_marking_s et	Sections 4.2 and 6
ART-LBL-06	ART-LBL-06_lithium_battery_ mark, and _reduced	REG-EEG-012 Rev B section 3.5
ART-LBL-07	ART-LBL-07_tamper_seal	Section 4.2
ART-PACK-01	ART-PACK-01_packing_layout_ provisional	Section 2.2 schedule and section 1.1 pocket column
ART-DIS-01	ART-DIS-01_disinfection_gui de_card	SVC-EEG-013 Rev B section 2 R5, which section 9 of that document names as this card's source

Identifier	File stem in graphics/labels/	Source of its content
ART-RET-01	ART-RET-01_return_shipping_instructions	Section 7, block quote
DRW-LBL-PLACEMENT	DRW-LBL-PLACEMENT_label_placement	Sections 4.1, 4.2, 3.2 and 6

Nothing in that table was typed into the generator. Every legend, size, substrate, placement and piece of body copy is read out of this document's own Markdown – or out of REG-EEG-012 and SVC-EEG-013 where those own the content – at generation time. Correct a legend here and re-run `python3 tools/artwork_gen.py`, and the tag follows; the artwork cannot drift from the document that owns it, which is the property section 2.3 claimed and had no mechanism to hold.

Four of the eleven cannot go to a printer as they stand, and say so on their own control sheet. ART-LBL-03, ART-LBL-05, ART-LBL-06 and ART-RET-01 carry <<PLACEHOLDER>> fields for the programme's postal address, its telephone number and its email address, none of which appears anywhere in package v2.3. The generator does not invent them: a wrong address on a case base is a kit that does not come back, and a dead telephone number on a regulated dangerous-goods mark is a parcel refused at a counter with a participant standing at it. ART-LBL-06 is incomplete for a second reason, in open item 14.

Two barcodes are encoded rather than pasted in. The ART-LBL-01 Data Matrix is a real ECC200 symbol – a 32 x 32 module symbol in the 12 x 12 mm this document specifies, so a 0.375 mm module, about 4.4 dots at 300 dpi – and the ART-LBL-02 kit id is real Code 128 subset B. Both encoders carry a self-test, run every time the artwork is generated: `python3 tools/artwork_gen.py --selftest` checks the Data Matrix against the ISO/IEC 16022 worked example and by decoding its own symbols, and checks the Code 128 pattern table structurally. **That proves the encoder, not a printed label.** Section 4.1 already requires ISO/IEC 15415 grade C or better after fifty wipes with 70 % IPA, and that first-article verification is what releases this artwork for use. Nothing here has been printed, applied, wiped or read by a verifier.

5. Serial number scheme

This section is the single home of the TI0V-B-nnnn serial format. It is defined here and nowhere else. FW-EEG-001 Rev D, WH-EEG-008 Rev B, ASM-EEG-007 Rev B, QP-EEG-010 Rev B, TST-EEG-004 Rev C, IFU-EEG-014 Rev B, PARTS-EEG-019 Rev B and RFQ-EEG-001 Rev E F-04 and M-03 all use the format and all **cite** this section for it rather than restating it.

Three other homes have been claimed for it and none of them is one. **FW-EEG-001 section 7 defines the public-key fingerprint, not the serial**, so a document citing FW-EEG-001 section 7 for the serial format has the wrong section of the wrong document. **PARTS-EEG-019 section 4.2 registers the three identifiers a unit carries** – the programme serial, the ATECC608B factory serial and the fingerprint – and cites this section for the format of the first of them; it settles which identifiers exist, not how the

serial is written. **RUL-EEG-021 Rev B section B** records the ruling that fixed the format; it is the minute of the decision, and this section is the specification the ruling asked for.

Format, fixed by this document:

TIOV-B-nnnn

example TIOV-B-0001

unit number, four digits, allocated in phase blocks

hardware revision letter of the carrier (EEG-CAR-01 Rev B)

programme prefix

Range	Reserved for
TIOV-B-0001 ... 0009	Phase 1. TIOV-B-0001 is the golden unit and is re-measured at every phase
TIOV-B-0010 ... 0099	Phase 2
TIOV-B-0100 ... 0999	Phase 3

Rebuilds take the next free number in their own phase block; there is no separate rebuild range. When the carrier goes to Rev C the revision letter in the serial changes with it, so a serial states which hardware it names without a lookup.

What was withdrawn, and why. Revision A of this document specified 0V-EEG-
<p><nnn><C> with a modulo-23 check character. That scheme is withdrawn entirely. Five forms were in circulation across the package – 0V-EEG-1001P here, 0V-EEG-
<phase><nnn> in ASM-EEG-007, TIOV-EEG-<phase>-<nnn> in QP-EEG-010, TIOV-EEG-0007 in WH-EEG-008 and TIOV-B-0007 in FW-EEG-001 – and the same string has to appear on the label, in the Data Matrix, in the USB iSerialNumber, in the calibration record and on this packing list. A check character that only one of six documents knows about is worse than no check character, and the firmware already writes TIOV-B-nnnn into the descriptor. One form, everywhere, wins. **CN-PKG-01** in the ECO-EEG-016 register records the withdrawal and requires ASM-EEG-007, QP-EEG-010, WH-EEG-008, TST-EEG-004 and JIG-EEG-009 to carry TIOV-B-nnnn at their next revision, including the golden unit, which is TIOV-B-0001 and not SN0001 or TIOV-EEG-1-001.

Proposed, and not yet ruled: TIOV-B-0000 is reserved for artwork specimens. The blocks above allocate 0001 to 0009, 0010 to 0099 and 0100 to 0999, so 0000 is in none of them. The specimen unit label of section 4.3 is drawn for TIOV-B-0000 precisely because of that: a specimen printed by accident, or a proof left on a bench, cannot be mistaken for the label of a unit that exists. Reserving it costs nothing and closes a hole that a specimen drawn for TIOV-B-0001 – the golden unit – would open. The programme owns the decision; it is open item 18 of section 9 until it is taken.

Serials are allocated by the programme in blocks and issued with the purchase order. A manufacturer never invents one. A unit that is re-boarded keeps its serial only if the ATECC608B moves with it; a new secure element means a new serial from the same phase block and a new device history record cross-referenced to the old one.

Three identifiers, one binding. The programme serial TIOV-B-nnnn above is printed, human-read, **and is what the firmware writes into the USB iSerialNumber** (RFQ F-04, and RUL-EEG-021 section B rules it). It is therefore what the browser’s persistent WebUSB authorisation binds to. The ATECC608B’s own 9-byte factory serial, rendered as 18 uppercase hex characters, is a *second* identifier: it is printed and carried in the

Data Matrix so a swapped secure element can be detected, and it is **not** the descriptor string. The fingerprint of section 4.1 is derived from the provisioned public key. All three are printed on the label, encoded in the Data Matrix, written into the calibration record at T5b and T6 and filed in the device history record. If any two disagree the unit is quarantined.

Kit id versus unit serial. KIT -<nnn> identifies the circulating asset – case, foam, helmet, consumables – and is printed on ART-LBL-02. The unit serial identifies the electronics. They are recorded as a pair at despatch and may be re-paired at refurbishment, for example when a helmet frame is withdrawn and the pod moves to a new frame. The pairing history lives in the device history record (QP-EEG-010).

6. Outer shipping carton

The travel case is **never** the shipping container (RFQ M-07). A case used as a parcel arrives scuffed, its latches sprung and its labels abraded, and it is a BOM item that has to survive twenty round trips.

Property	Value
Board	Double-wall BC flute, kraft
Strength	Box compression ≥ 7.0 kN; edge crush ≥ 1.9 kN/m
Internal size	Case external plus 40 mm on every face. 640 × 535 × 345 mm against the Peli 1560's published external of about 560 × 455 × 265 mm. Published, not measured: the carton is not ordered until the shell has been measured (section 3.2)
Void fill	Eight 40 mm PE foam corner blocks, 28 to 33 kg/m ³ . No loose fill – the case is smooth and slides
Closure	H-taped, 50 mm PP tape, plus one numbered security seal
Uses	Two: out and back. The participant re-uses the same carton and the same corner blocks
Label pocket	Transparent adhesive document wallet, 225 × 165 mm, on one long face, holding the pre-paid return label, the return instruction sheet and any customs paperwork (M-07)
Marking	ART-LBL-05 and ART-LBL-06 (section 4.2)
Spares	Two spare cartons and two spare label sets per ten kits, so a crushed carton does not strand a kit in a participant's home

Calculated masses. Nothing has been weighed. These figures are recalculated for the 150 × 130 mm four-layer carrier, the enlarged POD-P1 and the Rev C foam stack.

Item	Calculated mass	Basis
Helmet assembly (HM-01 240 g plus fitted parts)	500 g	Print volume $\times$ 1.01 g/cm ³
Pod assembly: POD-P1 base and lid 279 g, carrier 104 g, MP-01 48 g, thirteen module assemblies 130 g, cell 48 g, standoffs, screws and jumpers 37 g	650 g	Was 520 g in Rev A. The enclosure and the four-layer carrier account for the difference
Headphones	240 g	Published
Cables and charger	150 g	Published
Electrodes, ear clips, EMG leads	90 g	Estimated
Consumables pouch	450 g	Filled
Foam insert, Rev C stack, after cut-outs, plus the lid pad	about 600 g	516 $\times$ 390 $\times$ 175 mm is 35.2 litres of PE at 30 kg/m ³ , which is 1.06 kg solid; the openings of the seven Rev C cut files remove 15.8 litres, 45 % of it, leaving 583 g, and the 15 mm convoluted lid pad over the helmet bay adds about 20 g. Taken off the cut files of section 2.2 by planimeter with the corner radii included, at the nominal 30 kg/m ³ of the 28 to 33 kg/m ³ band, and not weighed . Rev A's 90 g was the two-layer sheet and is no longer relevant
Documents	60 g	A4 wallet
Packed contents	2.74 kg	
Case, empty	5.1 kg published, not weighed	The Peli 1560 without foam (section 3.2). The $\leq$ 4.0 kg target is withdrawn: no shell that meets the internal envelope is a 4.0 kg case
Gross packed case	7.8 kg	
Carton, corner blocks, tape and seal	0.9 kg	
Gross shipping mass	8.7 kg	

Volumetric mass, 640 × 535 × 345 mm at divisor 5000	23.6 kg	With the carton above
---	----------------	-----------------------

The consignment is charged on volume, not on mass: 23.6 kg volumetric against 8.7 kg actual. That is worth knowing before the fleet is sized. The case is the postage cost, and settling it on a shell large enough to stand the helmet upright took the chargeable weight from 17.4 kg to 23.6 kg, about 35 % more on every leg, out and back. The alternative was a case the kit does not fit in.

Drop rating, two separate tests. RFQ M-04 is a product test: 1 m onto concrete on each of six faces, case open and kit packed, on one Phase 1 unit, with pass defined as a re-run of TST-EEG-004 T3, T7, T8, T10 and T14 inside their original limits plus a cell inspection. Section 7's package test is a transport test on the sealed carton. Both are Phase 1 type tests, reported once per phase. Sign-off: programme, before Phase 2 is released. An ISTA 3A-equivalent transit sequence on one packed carton is run alongside them.

7. Lithium battery shipping

This section is the single home of the lithium-shipping procedure. REG-EEG-012 Rev B section 3 states the obligation and cites this section; TST-EEG-004 Rev C T29 verifies at kit closure against this section. None of the three restates the others.

Every kit contains one protected 18650 cell installed in the instrument. At 3.6 V nominal and 3400 to 3500 mAh that is **12.24 Wh (NCR18650B) or 12.6 Wh (Samsung 35E)** – calculated, under the 20 Wh per cell and 100 Wh per package thresholds. One cell, no battery pack.

The classification decision, recorded. DSN-EEG-002 Rev E section 7 said “a charged spare travels in the case, so a flat cell never ends a session”, and CASE-00 cuts a SPARE CELL pocket. A loose spare makes every consignment *packed with equipment* under packing instruction PI966, with different packing, marking and quantity rules, and it puts a bare lithium cell in a stranger's home. **The spare cell is deleted from the circulating kit.** The kit ships as UN 3481, lithium ion batteries **contained in equipment**, PI967 section II, on every leg. This is what RFQ-EEG-001 Rev E S-09 requires. The SPARE CELL bay stays in the foam and stays empty in circulation; it carries a spare only on depot-to-depot moves handled by the programme's trained shipper, and its tag reads SPARE CELL - - DEPOT ONLY, EMPTY IN CIRCULATION. The compensating control is RFQ E-22: at least four hours at 1000 Hz against a two-hour session, plus the charger of A-07.

Control	Value	Verification	Sign-off
UN number and packing instruction	UN 3481, PI967 section II	Consignment record	DG-trained shipper

Control	Value	Verification	Sign-off
State of charge at despatch	$\leq 30\%$	MAX17048 gauge read at the packing station through the session runner and written on the despatch record. TST-EEG-004 has no numbered step for this read – T24 is the radio-silence check and the Rev A citation of it here was wrong – and it is raised as open item 8 in section 9. Open item 8 is the only entry the package carries for this gap: RFQ-EEG-001 Rev E section 9.1 item 12 says the fuel-gauge reading has no per-unit step and is “raised as an open item” without naming a register, and neither RFQ-EEG-001 section 12 nor TST-EEG-004 section 16 carries one. Until one of them does, this is where the item lives	Packer
Cell secured	HM-10 keyed carrier behind the quarter-turn HM-08 hatch; cannot be inserted reversed (S-04)	Shake test at T18	Packer

Control	Value	Verification	Sign-off
Protection against accidental activation	The two mechanisms of RFQ S-01, and only those two: firmware refuses CMD_START_SESSION while VBUS_DET is high, and CHG_CE holds the charger disabled for the whole of a session. Rev A of this document also claimed a “hatch interlock ends a session”. No hatch interlock exists anywhere in the design and that claim is deleted	T4 and T21	Production QA
Outer packaging	Strong rigid outer, the section 6 carton	Visual	Packer
Package drop test	1.2 m, six orientations, on a completed package, with no cell damage, no leakage and no shifting of contents	Once per carton design, report filed	Programme
Lithium battery mark	ART-LBL-06 on the outer carton, with UN 3481 and a telephone number	Visual at despatch, and T29	Packer

The $\leq 30\%$ state of charge is a **programme control, not a regulatory requirement** for UN 3481 in equipment. It is applied because it lowers the energy in a damaged package and because S-09 requires it. Likewise, a single-kit parcel with one cell installed falls inside the small-package relief in PI967 section II and would not strictly need the mark; the programme applies the mark to **every** carton anyway, because a participant handing a parcel across a counter needs the parcel to answer the counter’s question for them. Consignments of **more than two packages** exceed the relief and require the mark regardless, which is the threshold REG-EEG-012 section 3.3’s configuration table states in its own words. That is the whole of the programme’s position on the mark, and the two documents state one rule.

Documents held per shipment. UN 38.3 test summary from the cell supplier, per cell lot, obtained before the first shipment and not after (S-04); cell safety data sheet; the package drop-test record; the consignment record; and, where the carrier requests it, the written statement that the consignment is UN3481 in compliance with section II of PI967. That list is owned by **REG-EEG-012 section 3.7** and this is the packer’s rendering of it. The programme puts the statement on every air waybill rather than waiting to be asked, in the words “Lithium ion batteries contained in equipment, UN 3481, PI967 section II, not restricted”. **That wording is not prescribed by the**

regulations and has not been checked against the edition in force; it is one of the things the DG-trained shipper verifies under REG-EEG-012 section 3.3 before each phase's despatch. No shipper's declaration for dangerous goods is required for this configuration. Sign-off: the programme's DG-trained shipper, named in the quality plan, with a stated refresher interval. The manufacturer answers the same question in the RFQ compliance matrix.

What a courier will and will not accept. Cells installed in equipment, one per parcel, declared, are accepted on road and on most air services by bpost, DPD, GLS, UPS and DHL. Loose cells on a consumer counter service are refused by most of them, which is a second reason the spare is gone. Any given carrier's consumer-return terms must be confirmed **in writing** before a return label is bought, and the confirmation filed. One primary and one alternate carrier are named per phase in the logistics file.

The return leg. The participant is legally the shipper and cannot discharge a shipper's obligation. The programme therefore does the work in advance: a pre-paid, pre-classified return label on a lithium-approved service is printed at despatch and travels in the carton's document pocket, together with a printed sheet the participant can hand over. The participant never writes a shipping document, never opens the pod and never handles a cell. The instruction sheet ART-RET-01 reads, in substance:

Repack the case to the photograph inside the lid, close both latches, put the case back in the same carton with the four foam blocks, tape it shut, and stick on the pre-paid label from the clear pocket. The label is already correct for the battery inside the instrument. If the counter asks what is inside, say: *a research instrument with a small rechargeable battery fitted inside it, UN 3481, already declared on the label*. Do not open the instrument, do not remove any battery, and do not post it in a letterbox – hand it over at a counter or book a collection. If the carton is damaged, telephone us and we will send another; do not use a different box.

Cells are stored at the depot at 40 to 60 % state of charge, never on charge unattended, in a metal-free box. A swollen, dropped or deep-discharged cell is quarantined against the kit id and replaced from spares, never shipped. End-of-life cells go to the Belgian collection scheme.

8. Despatch and receipt records

One record per leg, held against the kit id and the unit serial, in the device history record (QP-EEG-010).

Despatch record – completed by the packer, countersigned by a checker, before the carton is sealed:

Field	Source
Kit id, unit serial TI0V-B-nnnn, ATECC serial, fingerprint	Label and calibration record
KPL-EEG-001 ticked, both signatures	Section 1
Cell lot and state of charge at despatch	Gauge read, $\leq 30\%$
Case seal numbers, both hasps	ART-LBL-07

Field	Source
Carton security seal number	Section 6
Firmware version and image SHA-256	Provisioning record
Consumables lot numbers and expiry dates	Pouch
Gross shipping mass, measured	Scale
Carrier, service, tracking number, declared value, insured yes/no	Booking
Outbound label and return label tracking numbers	Booking
Lithium mark applied, document set present	TST-EEG-004 T29
Destination reference, not the participant's name	Study register
Packer, checker, date	Signature

Receipt record – completed at goods-in in Brussels, before the kit is opened for refurbishment:

Field	Check
Carton condition, crush or wet damage	Photograph if damaged
Carton seal number matches despatch	Yes / no
Case seal numbers match despatch	Yes / no. A broken seal means the case was opened in transit; quarantine
Photograph of the open case against the packing photograph	Filed
KPL-EEG-001 return column ticked	Every line
Missing or damaged items, with line numbers	Listed
Cell state of charge and gauge-reported capacity on return	Recorded
Quarantine 24 h before handling	Initialled
Reduced re-test result after refurbishment (SVC-EEG-013)	Values recorded
Deposit release authorised	Signature

A kit is not released back into circulation until the receipt record, the refurbishment record and the reduced re-test all carry a signature. Records are retained for the life of the programme plus five years, held by the programme, with a copy of the per-unit build

and calibration record supplied by the manufacturer as a condition of acceptance under RFQ-EEG-001 Rev E section 9.3.

9. Open items in this document

#	Item	Gate
1	The CASE-00 Rev C cut files are drawn – seven layer files, listed in section 2.2 – and nothing has been cut from them . Six of the nine bays hold parts that are not dimensioned anywhere in the package, so their plan sizes and every packer are provisional until the trial pack, and the sheet is drawn to a published case dimension rather than a measured one. Phase 1 packs in pluck foam	Blocks Phase 2 kitting
2	No travel case has been bought, measured or weighed. Section 3.2 settles the internal figures on AVL-EEG-017 K21's transcription of the Peli 1560, and the foam sheet is drawn to it with only 2 mm of margin on each plan axis, so a shell that measures short makes the cut files scrap. The base-to-lid depth split is not published and the ≥ 185 mm base figure is unverified	Blocks the foam cut , the carton order and the freight quote. The cut files themselves are drawn
3	RFQ-EEG-001 M-05's approximately $340 \times 250 \times 210$ mm internal is smaller than the $516 \times 390 \times 210$ mm requirement of section 3.2 derived from the bay schedule, and it is no longer restated anywhere else: AVL-EEG-017 K21 now cites section 3.2, so half of CN-PKG-03 is closed and RFQ-EEG-001 has still to be corrected at its next revision	Blocks a clean acceptance against M-05
4	Every mass, the volumetric mass and the carton size are calculated from geometry, published shell figures and assumed densities. Nothing has been weighed. The foam mass is now calculated bay by bay from the section 2.2 schedule rather than estimated, which makes it precise and still not measured	Closed by the first packed kit

#	Item	Gate
5	The UN 38.3 test summary has not been obtained from a cell supplier, because no cell has been bought	Blocks the first shipment, not the build
6	Carrier consumer-return terms for UN 3481 have not been confirmed in writing with any carrier	Blocks the first participant despatch
7	The packing photograph does not exist; a rendered provisional stands in	Closed at first build
8	The state-of-charge read at despatch has no numbered step in TST-EEG-004. It is performed at the packing station and recorded, but it is not a test step, and this document does not number one for it. This row is also the register entry for the wider gap: RFQ-EEG-001 Rev E section 9.1 item 12 states that the MAX17048 fuel gauge has no per-unit step and is “raised as an open item”, and no such item exists in RFQ-EEG-001 section 12 or in TST-EEG-004 section 16, so the reference points nowhere and this row is what it should point at	Closed when TST-EEG-004 adds a step or accepts the packing-station record as the evidence. The dangling reference is closed by RFQ-EEG-001 at its next revision
9	The host connector is a live non-conformance against RFQ E-24: the only named isolator module presents USB-B and the WH-09 pigtail is an interim answer. WH-09 is not yet in the PARTS-EEG-019 register	Blocks the panel aperture drawing and the Phase 2 module order
10	The settled shell has no drawn alternate. The Nanuk 960 of AVL-EEG-017 K21 clears the depth but is 32 mm short of the 390 mm second axis, so the CASE-00 Rev C bay layout does not fit it; using it means a Rev D of the cut file. A single-source case is a supply risk on a five-year programme	Closed by a second shell being laid out, or accepted as a risk by the programme
11	The ≤ 4.0 kg empty-case target of Rev A is withdrawn in section 3.2 because no shell that meets the internal envelope meets it. That is a requirement relaxed to fit the parts, and RFQ M-05 carries no empty-mass figure to correct it against	Closed when M-05 is revised under CN-PKG-03

#	Item	Gate
12	The finger reliefs of the Rev C schedule remove no foam. Every relief centre in the section 2.2 table sits at least 3 mm inside its own bay opening, so the ø22 mm circle falls within material the bay profile already removes; the cut files draw them there rather than differ from the schedule. They cannot be moved outwards – a straddling relief needs 11 mm beyond the bay wall and every web is 8 mm – so the fix is a scallop in the first uncut layer beneath each bay, or a re-columned sheet, and both are a Rev D	Decided at the trial pack, with the six undimensioned bays
13	Four of the eleven artwork files cannot be printed. ART-LBL-03, ART-LBL-05, ART-LBL-06 and ART-RET-01 carry <<PLACEHOLDER>> fields for the programme's postal address, telephone number and email address. None of the three appears anywhere in package v2.3, and the generator will not invent them. The files are drawn, dimensioned and complete apart from those fields	Programme lead, before the first label order. Blocks ART-LBL-03, ART-LBL-05, ART-LBL-06 and ART-RET-01 only
14	ART-LBL-06 is incomplete by design: the regulated pictogram is not supplied. The outer size, the reduced size, the hatched border, the UN3481 text and the telephone field are drawn to the figures REG-EEG-012 Rev B section 3.5 states. The battery-group-and-flame symbol belongs to the edition of ADR / IATA DGR in force, this package does not hold it, and a drawn-from-memory symbol is a mark that gets refused at a counter – which happens to a participant, not to the programme	The programme's DG-trained shipper, per REG-EEG-012 section 3.7, before the first despatch
15	The 60 x 12 mm bay tag cannot carry an 8 mm cap height, and this is measured. Section 4.2 fixes the tag and section 2.3 asks for the cap height; not one of the nine legends fits on one line, the least demanding needing about 88 mm of tag and the worst 320 mm. The generated tags are set between 5.36 and 2.68 mm on the specified tag. About 171 x 21 mm carries every legend at 8 mm on at most two lines	Programme, with the trial pack of open item 1. Changing the tag changes section 4.2 and section 2.3 together

#	Item	Gate
16	The spare-cell bay tag is specified twice and the two disagree. Section 4.2 makes the ART-LBL-04 legends identical to the section 2.2 schedule, which reads SPARE CELL; section 7 makes this one tag read SPARE CELL -- DEPOT ONLY, EMPTY IN CIRCULATION. The generated tag carries the section 7 string. One of the two sections has to change	Programme. Closed when this document is next re-issued
17	DRW-LBL-PLACEMENT's positions are a proposal, not a measurement. The faces and the label sizes are section 4.2's and are fixed; where each label sits on its face is the drawing's, against a case whose external dimensions are published rather than measured (open item 2) and a carton sized from those. Positions are dimensioned from the two nearest edges so that a measured shell changes the numbers and not the arrangement. The pressure-equalisation valve is a written constraint on the sheet and not a drawn keep-out, because its position on the bought shell is not known	Packer and QA at the first packed kit, against a shell that has been measured
18	TI0V-B-0000 is proposed as the artwork specimen serial and is not yet ruled. Section 5 sets out why. The specimen unit label is already drawn for it; if the programme rules otherwise the artwork is regenerated	Programme. Closed by a ruling either way